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Pioneering Big Data Fracture Calibration Framework: EDFM-AI For Field-Wise Hydraulic Fracture Network Quantification

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Abstract

This paper presents a scalable EDFM-AI framework for automated calibration of effective hydraulic fracture geometries across multiple shale gas well pads, marking the industry's first integration of big data fracture modeling with machine learning. The proposed workflow combines geological and petrophysical inputs with a star-boundary fracture generator and EDFM-coupled GPU-accelerated reservoir simulation. It further integrates an ESM DA-assisted proxy model to address high-dimensional uncertainties in fracture geometry, enabling rapid and accurate history matching. The methodology is applied to 23 horizontal wells across four shale gas pads, resulting in calibrated fracture half-lengths and heights ranging from 70–220 meters and 10–25 meters, respectively. Bottomhole pressure matches exceed 90% accuracy, and the 20-year gas EUR forecast quantifies the long term production potential range to be 0.09–0.15 billion cubic meters, showcasing the model's predictive capability. This novel workflow transforms unconventional reservoir characterization by enabling data-driven, field-scale fracture calibration and production forecasting at previously impractical scales. Its automation, scalability, and high fidelity make it a powerful tool for optimizing shale gas development across large well populations.

Introduction

Accurate integration of hydraulic fracture geometry into reservoir simulation remains one of the most technically demanding challenges in modeling unconventional reservoirs. Traditional simulation frameworks such as local grid refinement (LGR), dual-porosity-dual-permeability (DPDK) models, and unstructured grids often fall short when applied to the complex, irregular geometries typical of field-scale hydraulic fractures. LGR-based approaches, while conceptually straightforward, require dense gridding around fractures, leading to steep computational costs and convergence instability in large models (Tijink and Cottier, 2019). DPDK models, on the other hand, tend to oversimplify the fracture geometry by treating fractures as homogenized continua, which is inadequate for capturing the irregular, high-conductivity pathways of actual fracture networks (Yu and Sepehrnoori, 2018). Similarly, although unstructured grids offer more geometric flexibility, their widespread application is hindered by the complexities of mesh generation and loss of computational efficiency (Ding, 2019). These modeling paradigms are often limited to simple planar or orthogonal fracture configurations and fail to represent the spatial heterogeneity and anisotropy of real-world hydraulic fractures (Wu and Olson, 2016; Xu et al., 2017). As a result, the development of a robust and scalable approach to embed realistic fracture geometries into numerical simulators remains a critical unmet need in hydraulic fracturing studies.

While significant efforts have been made to model large-scale reservoirs, the majority of these efforts have been focused on conventional reservoirs with relatively straightforward geological features and flow regimes (Gao et al., 2017). For unconventional shale reservoirs, the situation is markedly different: reservoir heterogeneity, complex geomechanical behavior, and large-scale fracture propagation make accurate modeling highly complex and computationally prohibitive. Existing large-model frameworks rarely consider the explicit geometry of hydraulic fractures due to their scale incompatibility with the computational costs of conventional fracture modeling methods. This mismatch has led to a prominent lack of research and practical tools tailored for full-field simulation of hydraulically fractured reservoirs. Moreover, many of the methods that do attempt to capture complex fractures are constrained to single-stage/single-well or small-pilot domains, limiting their scalability and predictive capabilities for reservoir-wide performance evaluation (McClure et al., 2020; Liang et al., 2022). As unconventional field developments increasingly rely on full-field design&optimization, there is a pressing need for a simulation framework that not only incorporates complex fracture geometry but also scales efficiently across hundreds of wells and kilometers of subsurface domains.

This paper presents a pioneering big data fracture calibration framework that leverages the Embedded Discrete Fracture Model (EDFM) to enable efficient, scalable simulation of field-scale unconventional reservoirs with complex hydraulic fracture networks. EDFM allows for the incorporation of arbitrarily shaped fractures without the need for intensive local grid refinement, significantly reducing the computational burden while preserving accuracy (Sepehrnoori et al., 2020; Xu and Sepehrnoori, 2019). By combining this modeling flexibility with AI-driven parameter calibration, the proposed framework represents the first application of EDFM to an actual field-scale shale gas case study, demonstrating its practicality and performance at unprecedented scales.

Methodology

The core simulation engine in this study is built upon the EDFM, a powerful and flexible framework for modeling complex fracture networks within conventional matrix grids. EDFM circumvents the limitations of conventional local grid refinement by introducing non-neighboring connections (NNCs) to calculate transmissibility between matrix blocks and fracture segments, as well as among intersecting fractures themselves. These NNCs enable the decoupling of fracture geometry from the underlying grid structure, allowing for the representation of arbitrarily oriented and irregularly shaped fractures without requiring gridding modifications. To address the scale and efficiency demands of field-wide simulation, EDFM

is implemented on a GPU-accelerated reservoir simulation platform. The parallelism inherent in GPUs significantly reduces computational time, especially in simulations involving thousands of fracture segments and millions of grid cells. Benchmark tests conducted on several fracture-dominated reservoir cases show that the GPU-based solver produces results that are virtually identical to traditional CPU-based simulators in terms of pressure, saturation, and production profiles, while achieving a 10~50 times speedup in run time depending on model complexity and fracture density.

To enhance model calibration and uncertainty quantification, this study integrates the EDFM& GPU reservoir simulation engine into a novel AI-driven workflow termed EDFM-AI, which leverages the Ensemble Smoother with Multiple Data Assimilation (ESMDA) algorithm. The EDFM-AI framework consists of four major components: (1) a parametric star-like fracture geometry generator; (2) automated EDFM preprocessing for NNC construction; (3) GPU-based reservoir simulation; and (4) an ensemble-based AI optimizer powered by ESMDA. The uniqueness of this workflow lies in its scalability to field-scale models with high-dimensional uncertainty spaces, including effective fracture dimensions, fracture conductivity, fracture water saturation, matrix permeability, etc. ESMDA, known for its robustness in high-dimensional inverse problems, is particularly well suited for unconventional reservoirs where fracture properties are often unobservable and heterogeneous. By iteratively updating the ensemble of model realizations to match historical production data, EDFM-AI efficiently converges to plausible fracture realizations that honor field observations. See Fig. 1 for the EDFM-AI workflow with ESMDA optimizer.

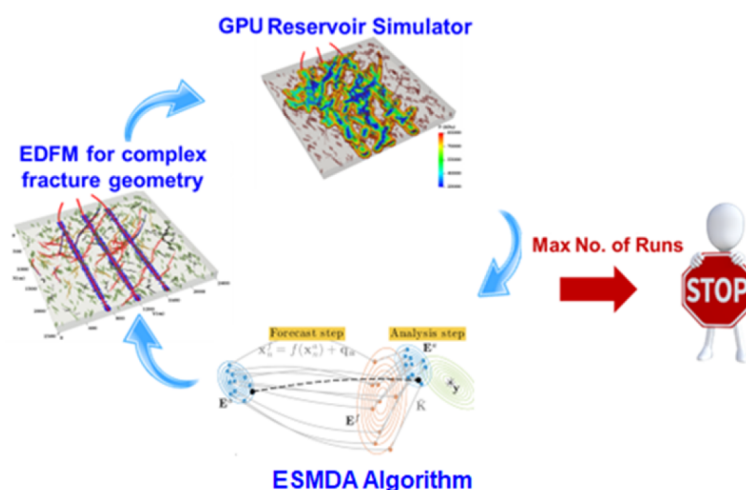


Figure 1—EDFM-AI workflow schematics combining EDFM, GPU-reservoir simulation, and ESMDA algorithm.

A key innovation in this study is a specialized hydraulic fracture generation module designed to produce star-like fracture geometries. Unlike conventional planar or rectangular representations, the star-shaped fracture design mimics field observations of radial fracture propagation in shale formations. The algorithm initiates hydraulic fractures at the perforation clusters with maximum vertical height and imposes an exponential decay of height away from the wellbore, capturing the physical tendency of fracture growth to taper with distance due to stress barriers and energy dissipation. This design ensures that the simulated fracture networks reflect both mechanical realism and operational pumping constraints. The geometric parameters of these fractures—such as half-wing length and maximum height—are randomized within defined ranges and treated as uncertain variables within the ESMDA-assisted history matching loop.

Field-Scale Application

The reservoir simulation model is constructed for a large-scale unconventional shale gas field spanning approximately 33.7 square kilometers, with a lateral extent of 6.3 km by 5.35 km and an average thickness

of 43 meters. The simulation domain was discretized using a finely gridded 3D model consisting of $214 \times 187 \times 68$ grid blocks, totaling over 2.72 million cells to capture the heterogeneous geological and petrophysical features. The grid resolution (size) in x, y, and z directions are 30 meters $\times$ 30 meters $\times$ 0.6 meters respectively. The reservoir lies at depths ranging from 1760 to 2570 meters and exhibits a highly flow-restrictive matrix system, with porosity varying from 3% to 7% and matrix permeability on the order of 200 nano-Darcy, representative of deep, tight shale formations. Four horizontal well pads were included in the study, encompassing a total of 23 laterals with approximately 1901 total completion perforation points distributed across all wells. The horizontal lateral length of the wells ranges from a minimum of 1250 meters to 2150 meters, with average lateral length of 1800 meters. The well spacing of all these four pads follows 500-meters design. The model incorporated up to 9 years of production history, representative of a well-developed field, to ensure robust calibration. Initial reservoir pressures range between 35 and 64 MPa, which is much lower compared to other blocks in the same formation that are much deeper and have much higher pore pressures. The key properties of the reservoir model are tabulated in Table 1, and the 3D visualizations of the reservoir properties/well trajectories are plotted in Fig. 2, Fig. 3, Fig. 4, and Fig. 5. This simulation framework was designed to integrate hydraulic fracture geometries, production constraints, and pressure depletion dynamics, forming the foundation for high-resolution history matching and EUR forecasting. This comprehensive model serves as the backbone for GPU-accelerated EDFM-AI simulations aimed at understanding field-scale fracture effectiveness and optimizing development strategies.

Table 1—Key reservoir, fluid and fracture properties used in the field-scale gas-condensate model.

Model parameter	Value	Unit
Reservoir area	33.7	square kilometers
Reservoir dimensions ($x \times y \times z$)	$6300 \times 5300 \times 43$	meters
Number of grid blocks ($x \times y \times z$)	$214 \times 187 \times 68$	meters
Size of grid blocks ($x \times y \times z$)	$30 \times 30 \times 0.6$	meters
Total number of grid blocks	2.72	million
Depth range of the reservoir	1760~2570	meters
Reservoir pressure range	35~64	mega-Pascal
Matrix porosity range	3~7	%
Approximate matrix permeability	200	nano-darcy
Number of pads/horizontal wells	4 / 23	-
Range/average of horizontal lateral length	1250~2150 / 1800	meters
Well spacings design	500	meters
Number of perforations (hydraulic fractures)	1901	clusters
Number of production history	9	years

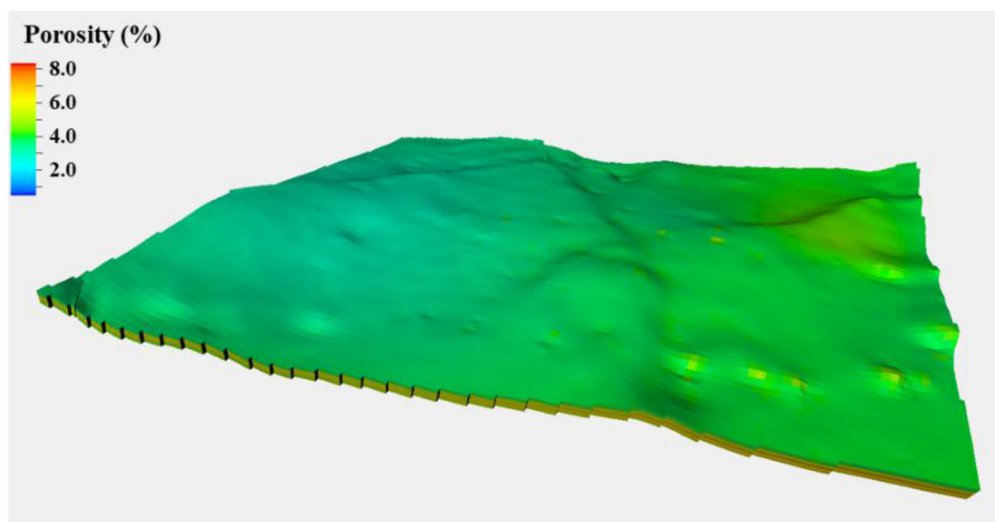


Figure 2—The field scale shale gas reservoir's porosity property distribution.

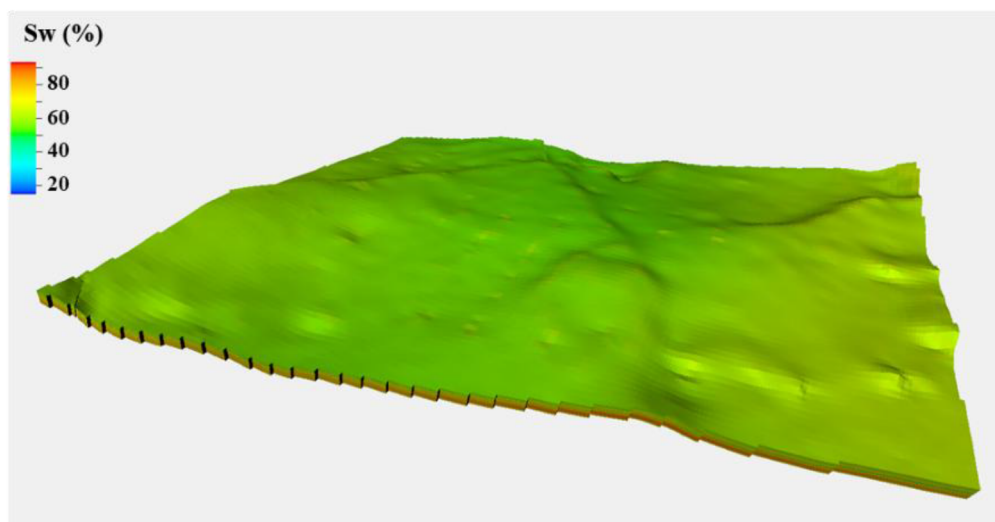


Figure 3—The field scale shale gas reservoir's water saturation property distribution.

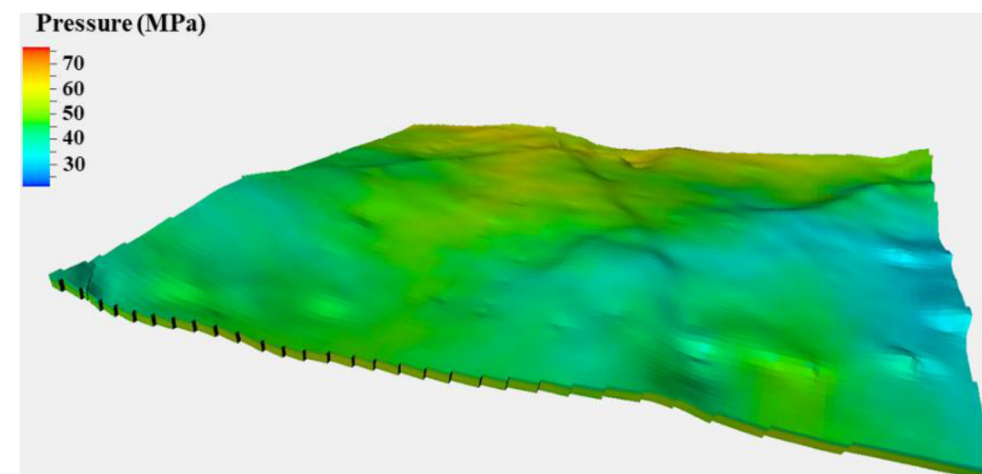


Figure 4—The field scale shale gas reservoir's initial reservoir pressure property distribution.

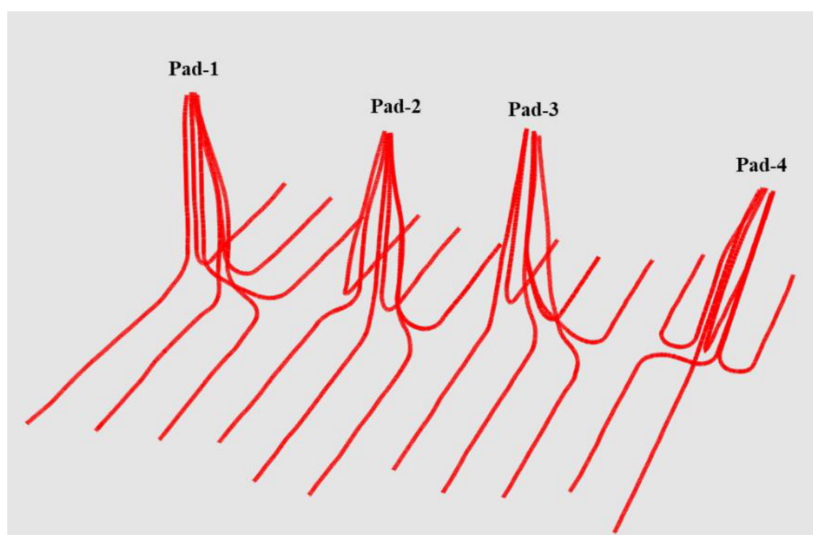


Figure 5—The well trajectories of the 4 pads, totaling 23 wells considered in this field scale model.

The hydraulic fracture geometry in this study is constructed based on field-specific geomechanical insights into the shale gas formation. As illustrated in the schematic shown on the left of Fig. 6, the fracture height is maximal at the point of perforation, where the fluid energy is mostly concentrated, and gradually diminishes toward the fracture tips. This tapering effect is driven by a combination of energy dissipation along the fracture wings and increasing resistance from in-situ stresses, particularly the horizontal stress gradient. The resulting fracture-plane's shape follows a star-boundary configuration, consistent with empirical observations and expert interpretations of fracture propagation behavior in this reservoir. This unique shape reflects the anisotropic nature of stress fields and rock properties, with a centrally peaked fracture height and a symmetric, tapered geometry extending outward. The figure on the right of Fig. 6 presents a 3D visualization of the generated hydraulic fractures from the fracture generator that will be fed into the reservoir simulation model, capturing the spatial extent and variability of the fracture network across the wellbore. This model captures the spatial extent and variability of the fracture network across the wellbore, and it aligns with the local geological and geomechanical understandings, ensuring a representative fracture geometry for the history matching process. Fig. 7 shows the final calibrated hydraulic fracture geometry after the history matching workflow. For the field-scale reservoir model, there are a lot of variabilities in the dipping angles of the formation, even along a single horizontal wellbore's lateral. The generated hydraulic fractures also have matching dipping angles of the formation, guaranteeing that no parts of fracture are exposed outside of the geology model. The ranges of effective fracture half-length and fracture height are 70~220 meters and 10~25 meters, respectively.

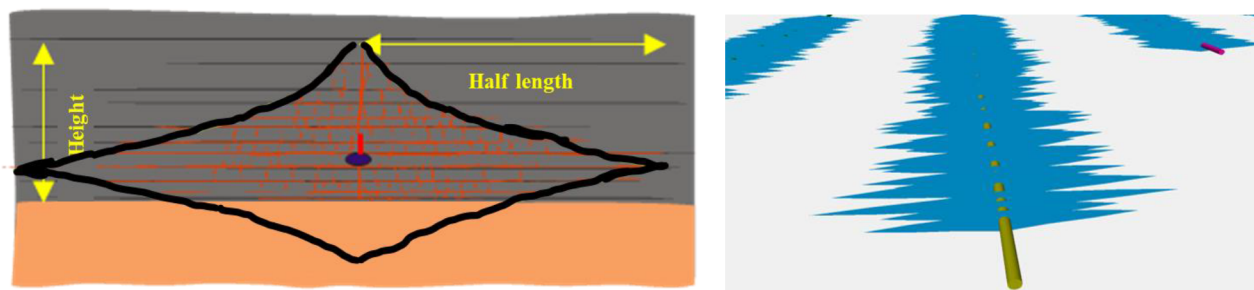


Figure 6—The cross sectional view of the theoretical hydraulic fracture geometry (left), and the generated hydraulic fractures model of one horizontal wellbore for reservoir simulation (right).

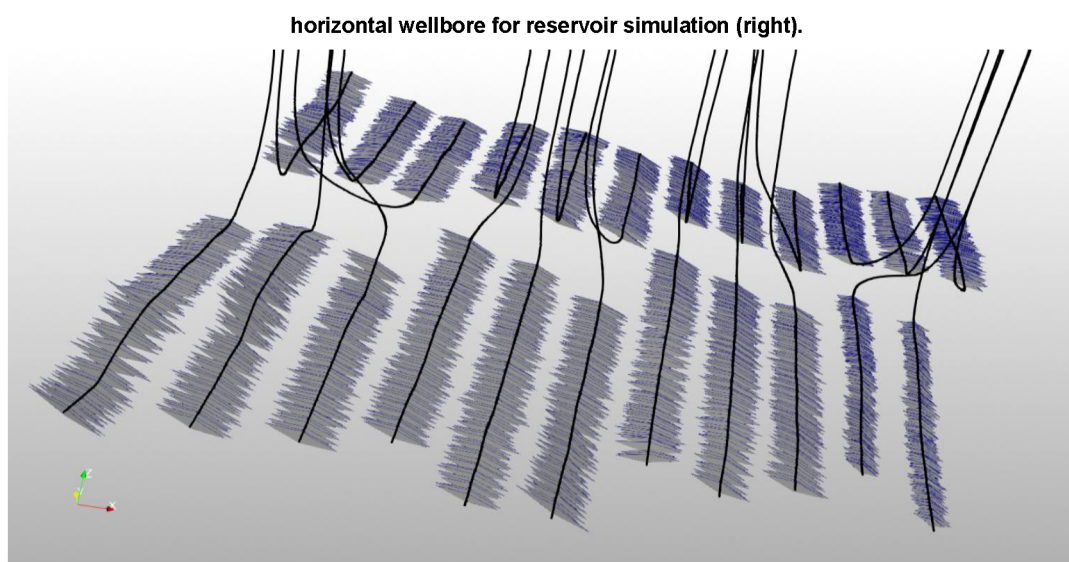


Figure 7—The final, calibrated effective hydraulic fracture model for this field-scale shale gas reservoir.

The bottomhole pressure (BHP) history matching results for six wells across three representative shale gas pads are illustrated in Fig. 8, showcasing the robust performance of the EDFM-AI integrated reservoir simulation framework. The history matching is done by using real gas flow rates as model constraint, while matching the BHP. 200 realizations of reservoir simulations are performed, and the best representative model's results are displayed. In each subfigure (a–c), the orange crosses denote the actual BHP history data, while the solid orange curves represent the simulation-calibrated results. The y-axis is the BHP with units of absolute bars (BARSA, equal to mega-Pascal [MPa] times 10), while the x-axis is the time dimension. As evident from the plots, most BHP measurements are limited to approximately the first five years of production, beyond which field data become unavailable. However, according to operational feedback and long-term field surveillance, the bottomhole pressures typically bottomout after year five, with no notable fluctuations, justifying the use of constant BHP constraints beyond the available history. The simulation demonstrates excellent agreements with observed data, achieving an average relative error of less than 10%. On top of this, one realization of the EDFM fracture preprocessing/GPU-reservoir simulation only takes roughly 2.5/13.5 minutes to finish (totaling 16 minutes). All six wells exhibit a steep decline in BHP during the early production stage, typically within the first one to three years, consistent with typical shale gas reservoir depletion behavior. The calibrated results reveal that the average stabilized BHP reaches a minimum of approximately 5~10 MPa. This history matching outcome validates the robustness of the fracture calibration process and confirms the ability of the EDFM-AI workflow to replicate complex, multi-well pressure dynamics at field scale.

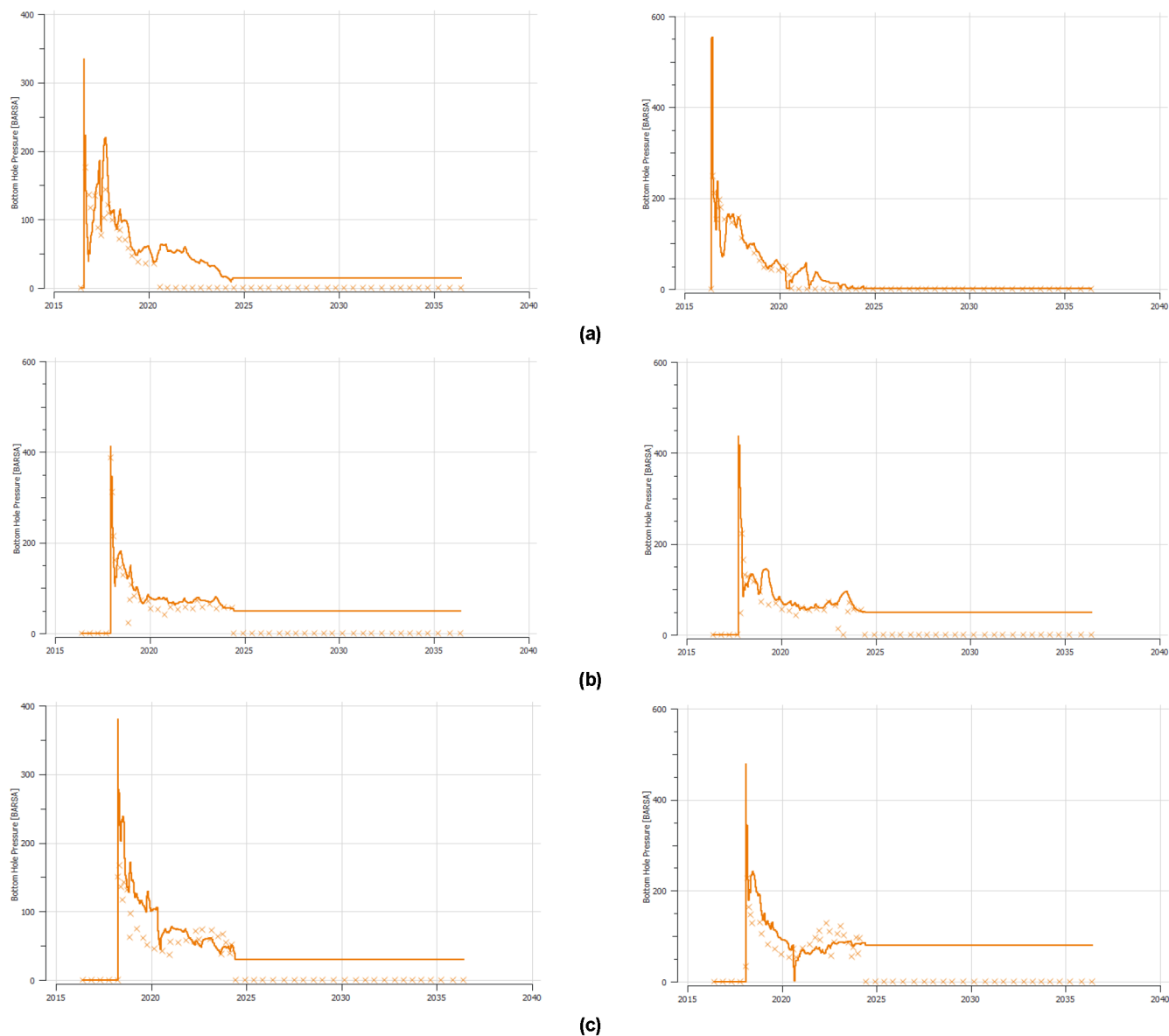


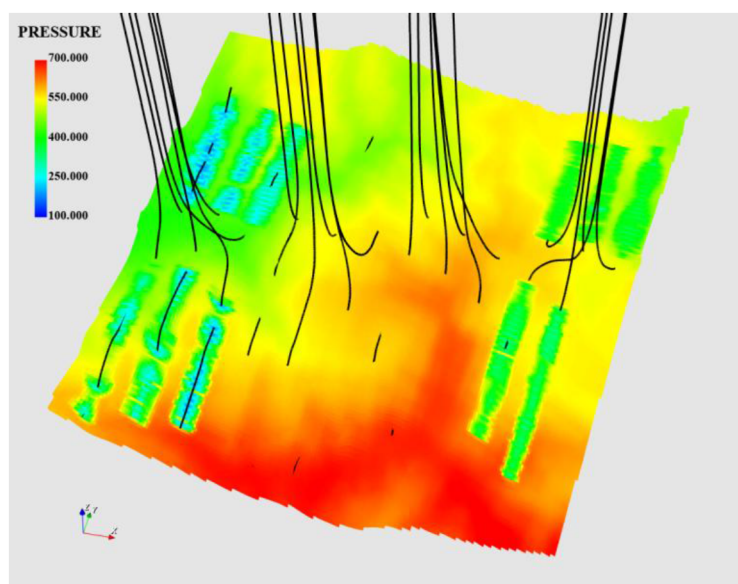
Figure 8—History matching results of the bottomhole pressures (BHP) of 3 selected pad, totaling 6 wells: (a) Pad-1, Well-3 (left) and Well-5 (right); (b) Pad-2, Well-1 (left) and Well-2 (right); (c) Pad-3, Well-2 (left) and Well-4 (right).

The 20-year production forecast results for all 23 horizontal shale gas wells are then simulated based on the best match model (results tabulated in Table 2). The forecasts were simulated using the BHP constraint to predict the projected cumulative gas production, which is opposite to the setting during history matching (gas rate as constraint, match the BHP). The estimated ultimate recovery (EUR) values span a range from 0.09 to 0.15 billion cubic meters across all wells, indicating notable well-by-well variability in reservoir quality, fracture effectiveness, and drainage efficiency. When the EURs are averaged by pad, clear differences can be observed: Pad-1 exhibits the strongest performance with an average EUR of 0.138 billion cubic meters (Bcm), followed by Pad-4 at 0.126 Bcm, Pad-2 at 0.121 Bcm, and Pad-3 at 0.1 Bcm. These differences may be attributed to geological heterogeneity, completion design, and variability in stimulated reservoir volume. The results offer valuable insights into pad-level performance, informing future infill planning, refracturing decisions, and economic optimization of shale gas development across this field.

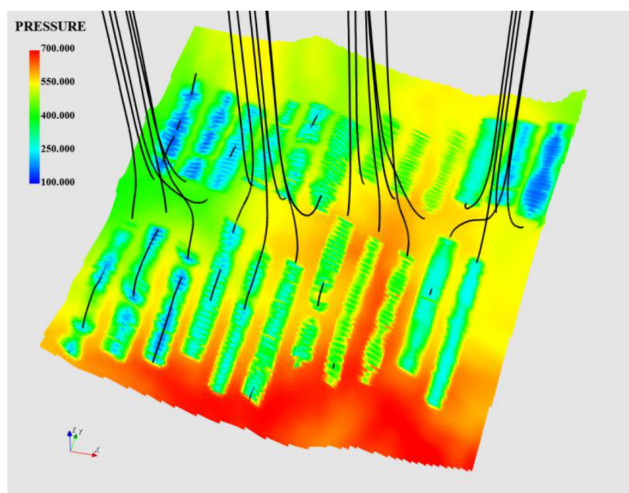
Table 2—Quantified 20-year EUR of the 23 horizontal shale gas wells of this field-scale reservoir.

Well	EUR value (Bcm)	Well	EUR value (Bcm)
Pad-1, Well-1	1.31	Pad-3, Well-1	0.92
Pad-1, Well-2	1.39	Pad-3, Well-2	0.90
Pad-1, Well-3	1.35	Pad-3, Well-3	1.16
Pad-1, Well-4	1.36	Pad-3, Well-4	0.90
Pad-1, Well-5	1.51	Pad-3, Well-5	0.92
Pad-1, Well-6	1.36	Pad-3, Well-6	1.21
Pad-2, Well-1	1.22	Pad-4, Well-1	1.43
Pad-2, Well-2	1.23	Pad-4, Well-2	1.27
Pad-2, Well-3	1.12	Pad-4, Well-3	1.15
Pad-2, Well-4	1.16	Pad-4, Well-4	1.18
Pad-2, Well-5	1.18	Pad-4, Well-5	1.27
Pad-2, Well-6	1.37	23-well average	1.21

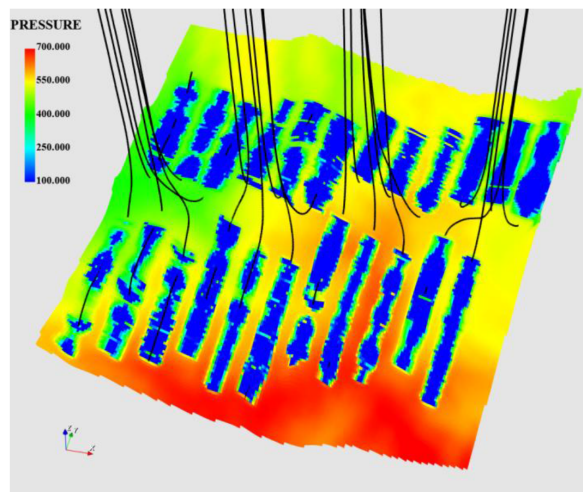
Fig. 9 shows the pressure drainage distributions of the main target layer after producing for 1 year, 2 years, 10 years, and 20 years respectively. From this figure, we can first observe from Fig. 9a and Fig. 9b that the drainage effects' onset happens earlier for Pad-1 and Pad-4 than for Pad-2 and Pad-3 since Pad-1/4 were put online earlier. If we compare the pressure drainage after 1 or 2 years, and the drainage after 10 or 20 years, it is concluded that the majority of the drainage happens in the early time, caused by the rapid hydraulic fracture closure. No significant drainage variations are seen between 10 years and 20 years of production. Finally, due to geological variations and obvious geomechanical/petrophysical/mineral content heterogeneity, the well trajectories are not always landed in the target layer, and the average landing precision is about 70% only. Therefore, significant heterogeneity is observed for pressure drainage after 20 years, even along a single horizontal wellbore.



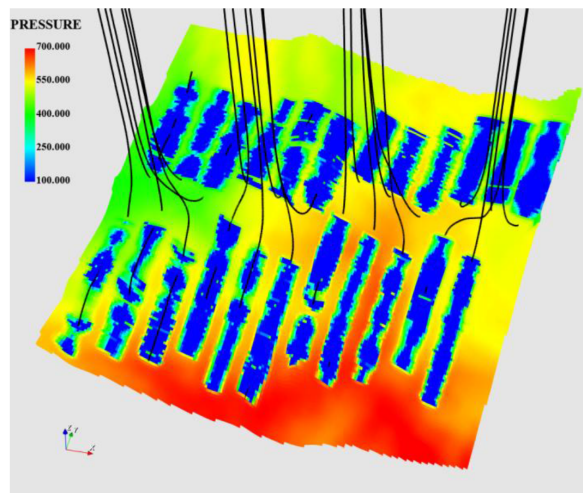
(a)



(b)



(c)



(d)

Figure 9—Dynamic pressure drainage of 23 wells' hydraulic fractures, for the field-scale shale gas reservoir at different producing time: (a) After 1 year; (b) After 2 years; (c) After 10 years; (d) After 20 years.

Conclusions

This work demonstrates a transformative advancement in unconventional reservoir simulation by enabling large-scale fracture quantification, reservoir model calibration, and production forecasting that were previously impractical due to computational limitations and challenges of numerical simulation of fractures. The EDFM-AI workflow is poised to serve as a scalable decision-support tool for the next generation of intelligent reservoir development strategies. The key points of this paper can be summarized below:

- A novel EDFM-AI framework was developed and applied to a large-scale shale gas reservoir model covering over 33 km² and 23 horizontal wells. This workflow integrates star-boundary fracture generation, EDFM, GPU-accelerated reservoir simulation, and ESMDA-based data assimilation, enabling efficient and automated field-scale fracture calibration.
- Quantitative evaluation of hydraulic fracture geometries was achieved, with effective half-lengths ranging from 70~220 meters and fracture heights from 10~25 meters. These calibrated values reflect significant heterogeneity across the field and provide more realistic representations compared to traditional planar, uniform fracture assumptions.
- Robust history matching results were achieved, with average bottomhole pressure matching accuracy exceeding 90%. The average simulation time per realization was only 16 minutes, demonstrating exceptional computational efficiency from the EDFM-GPU fractured reservoir simulation framework.
- The 20-year production forecasts confirmed the model's predictive capabilities, showing EURs between 0.09–0.15 billion cubic meters. Pad-1 outperformed others with an average EUR of 0.138 Bcm, underscoring the importance of field-scale modeling in optimization of future development strategies.

Acknowledgements

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